DE–6407 **101**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Comment the binary number 110111.110 in to octal and decimal number systems.

2. State and prove De Morgan's theorems.

3. Draw the circuit of a 4 × 1 multiplexer.

4. Draw the circuit of a half adder and give its truth table.

5. Explain the working of RS flip flop.

6. Explain any one secondary storage device.

7. What is the purpose of fetch cycle? Explain.

8. Explain the interrupt mechanism.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Subtract 23 from 82 by using 1's complement and 2's complement method.

10. Simplify by using Karnaugh map and development by using NAND gates only

$Q(A,B,C,D) = \sum m(2,3,4,7,8,12,14,15)$

11. Draw the circuit of a parallel binary adder and explain.

12. Draw the circuit of a 4-bit shift register and explain.

13. How many 256 × 8 RAM chips are needed to construct a memory of size 1KB? How will you connect it? Explain.

14. Explain the general register organisation with a block diagram.

15. Explain:

    (a) Programmed I/O

    (b) Decoder

    (c) Ripple counter.

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DE–6408 **102**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013

First Semester

C AND DATA STRUCTURES

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly explain how to close and open a data file in C.

2. Explain the conditional statement with an example.

3. Illustrate any two looping structure in C.

4. List out the advantages of using file.

5. Explain the various modes to open a file.

6. Write short notes on Trees and their applications.

7. Briefly explain about double linked list.

8. Write short notes on Dynamic memory allocation.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the data types in C language with an example.

10. Explain in detail about the Operators.

11. Write a C program to store and manipulate the employer records using structure.

12. Discuss the various operations on single linked list.

13. Write a C program to implement Macros.

14. Explain in detail about Binary tree and its traversals.

15. Discuss in detail about Multi dimensional array.

DE–6409 **103**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain schema and subschema with an example.

2. Write a note on 'Procedural and non procedural interfaces'.

3. What is meant by 'informating bearing'?

4. Describe the database objects defined by the oracle system.

5. Describe the internal data types in oracle.

6. What is meant by 'embedded SQL'?

7. What is the importance of oracle precompiler?

8. How will you work with 'NULL' values? Give an example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is DBMS? Discuss the advantages and limitations.

10. Discuss the Data Definition, Data manipulation and data control facilities offered by Relational model.

11. Elaborate the basic concepts of Relational model of Database.

12. Discuss the architecture of oracle.

13. Explain the following:

    (a) Oracle aggregate functions

    (b) Oracle expressions.

14. Explain the complete structure of SQL queries with atleast two examples.

15. Write the statements and explain with examples of the following:

    (a) Creating Table

    (b) Modifying Table

    (c) Renaming Table

    (d) Copying Table

    (e) Droping Table

DE–6410 **104**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

OPERATING SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write about the evolution of operating system.
2. Write how operating system acts as a resource manager.
3. Explain the issues of interprocess communication.
4. How dead lock can be avoided? Discuss.
5. Discuss the paging model of logical and physical memory.
6. How does a page fault can be removed? Discuss.
7. Explain the types of security in file system.
8. List out the information elements of a file directory.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the structure of an operating system.
10. Describes the working principles of time sharing systems.

11. Develop the Dekker's algorithm for solving critical section problem.

12. Explain any two processor scheduling algorithm with illustration.

13. Give an account on interrupt handler.

14. List out and explain the characteristics of simple paging and simple segmentation techniques.

15. List and explain any three file allocation methods in detail.

DE–6411 **201**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Explain the polymorphism. Give an example.

2. Write a C++ program to generate the prime numbers from 1 to n.

3. What is meant by the control structure? Explain any one control structure.

4. Write a C++ program to solve the all possible roots in a quadratic equations.

5. Explain the call by reference with an example.

6. How to overloaded functions? Give an example.

7. Explain the single inheritance with an example.

8. How to open and close a file? Give an example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write short notes on :

   (a) OOP paradigm

   (b) Benefits of OOP

   (c) Applications of OOP.

10. Explain the Hierarchy of C++ data types.

11. Write a C++ program to multiply any two order matrix with pointers.

12. Write short notes on :

   (a) Friendly functions

   (b) Inline functions

   (c) Operator overloading.

13. Explain the hierarchical inheritance with an example program.

14. (a) Explain the error handling during file operations.

   (b) Explain the command line arguments.

15. Write a C++ program to read the file 1 for the item name and cost and get the input as number of items and display the item name, cost, number of items and amount.

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DE–6412 **202**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Explain briefly about Sed editor commands with its various options and examples.

2. Explain about Switch-Case statement in shell programming with examples.

3. Explain the filters in Unix with examples.

4. Explain Unix system calls.

5. Write a shell program to print the given numbers is prime or not.

6. How to access the environment variables in Unix?

7. Write short notes on Standard input and output in Unix.

8. Write a shell program to implement four function calculator.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. How can you produce complex tables with tbl command?

10. Discuss the command line structure also explain the creation of new command in shell programming with eg.

11. Explain pattern scanning and processing language using awk filters available in UNIX.

12. Write short notes on the following :

    (a) Unix system calls

    (b) Directories and inodes.

13. Explain the steps in program development and document preparation.

14. Explain about features of macro package.

15. Explain briefly about Sed editor commands with its various options and examples.

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DE–6413 **203**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

COMMUNICATION SKILLS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a short note on agenda writing.
2. What are the barriers to communication?
3. How can a telephone be used effectively?
4. What are the answering skills?
5. How will you control a discussion?
6. How is body language a powerful communication?
7. Write a short note on mock negotiation.
8. How will you make use of notes and outlines?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the reading techniques.
10. What are the various types of communication?
11. How will you plan for an interview?

12. What are the participation techniques?

13. How will you manage the process of communication?

14. Explain the various types of presentation.

15. Discuss the techniques for presentation.

DE–6414 **204**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

COMPUTER GRAPHICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Give a brief account on hard copy devices.

2. Write about the working principle of Light pen device.

3. Discuss the concept of Sutherland-Hodgeman polygon clipping method.

4. Briefly explain the procedure of window-to-viewport transformation principle.

5. With matrix representation, explain the three dimensional mirror reflection method.

6. Give a sketch on clipping against a finite view volume.

7. Elucidate the three dimensional viewing technique.

8. Write about components of user interface.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR of the following questions.

9. With neat diagram, explain the working principle of a Cathode Ray Tube.

10. Describe the procedure of circle generating algorithm with program segment.

11. List and explain the basic two dimensional transformation principles.

12. Narrate the Cohen-Sutherland line clipping algorithm.

13. Discuss on three dimensional rotation transformation.

14. Elaborate on hidden surface elimination with algorithm.

15. Give a neat description on various styles of command language.

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DE–6415 **301**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Show that $(P \to (Q \to R)) \Rightarrow (P \to Q) \to (P \to R)$.

2. Demonstrate that R is a valid inference from the premises $P \to Q, Q \to R$ and P.

3. Define the following with an example

   (a) subset

   (b) equal sets

   (c) proper subset

4. If Z be the set of integers and if R be the relation called "Congruence modulo 3" defined by

   $R = \{< x, y > / x \in Z \wedge y \in Z \wedge (x - y) \text{ is divisible by } 3\}$

   Determine the equivalence classes generated by the elements of Z.

5. If $f : R \to R$ be given by $f(x) = x^3 - 2$. Find f^{-1}.

6. For any commutative monoid $\langle M,* \rangle$, prove that the set of idempotent elements of M forms a submonoid.

7. If $\langle H,* \rangle$ be a subgroup of $\langle G,* \rangle$. Prove that the set of left cosets of H in G form a partition of G.

8. Define the following with an example :

   (a) graph

   (b) adjacency matrix of a graph

   (c) multiple graph

   (d) sub graph.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Obtain the principle conjunctive normal form of $(\neg P \vee \neg Q) \rightarrow (P \rightleftarrows \neg Q)$.

   (b) Obtain the principal disjunctive normal form of $P \vee (\neg P \rightarrow (Q \vee (\neg Q \rightarrow R)))$. (8 + 7)

10. (a) Show that $P(x) \wedge (x)Q(x) \Rightarrow (\exists x)(P(x) \wedge Q(x))$.

    (b) Construct the derivation of the equivalence $P \rightarrow (\exists x)Q(x) \Leftrightarrow (\exists x)(P \rightarrow Q(x))$ – without using E_{32}. (8 + 7)

11. Given $S = \{1,2,.....,10\}$ and a relation R on S where $R = \{< x, y > / x + y = 10\}$. What are the properties of the relation R?

12. Write about Hashing functions with an example.

13. (a) If $\langle S,* \rangle$ be a given semigroup. Then prove that there exists a homomorphism $g : S \to S^s$, where $\langle S^s, \circ \rangle$ is a semigroup of functions from S to S under the operation of (left) composition.

(b) Show that in a group $\langle G,* \rangle$, if for any $a,b \in G$, $(a * b)^2 = a^2 * b^2$, where $\langle G,* \rangle$ must be abelian. (8 + 7)

14. (a) If g be a homomorphism from a group $\langle G,* \rangle$ to a group $\langle H, \Delta \rangle$ and if K be the Kernel of g and $H' \subseteq H$ be the image set of g in H. Then prove that G/K is isomorphic to H'.

(b) Prove that the ring of even integers is a subring of the ring of integers. (10 + 5)

15. (a) In a simple digraph, prove that the length of any elementary path is less than or equal to $n-1$, where n is the number of nodes in the graph.

(b) Write a short note about the following with an example.

(i) Directed tree

(ii) Binary tree. (8 + 7)

DE–6416 **302**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

COMPUTER NETWORKS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various uses of computer network.

2. Explain the various guided transmission media.

3. Explain briefly on petrinet models.

4. Explain briefly about collision-free protocols.

5. Compare and contrast the datagram and virtual circuit subnets.

6. Write short notes on tunneling.

7. Explain about crash recovery in transport layer.

8. Explain the basic functions of e-mail.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. With a neat diagram, explain the functions of layers in OSI reference model.

10. Describe in detail, the public switched telephone network.

11. Explain with suitable examples, the error detection and correcting codes.

12. Explain with an illustration, the shortest path routing algorithm.

13. Discuss in detail on firewalls.

14. How do you measure the performance of a network discuss?

15. Discuss in detail on World Wide Web.

DE–6417 **303**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

SOFTWARE ENGINEERING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define Software Engineering. Explain the various characteristics of software.

2. Describe the spiral model of software development.

3. Describe briefly about programming team structure.

4. How do you estimate software maintenance cost? Explain.

5. Describe briefly about software scheduling plan.

6. Discuss the various risks in the software.

7. What is meant by software prototyping? Explain.

8. Write short notes on unit testing.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain in detail the phased model of software development.

10. Describe the COCOMO cost estimation model.

11. Discuss in detail on software configuration management.

12. Discuss in detail on software requirement specification.

13. Explain the various design principles used for software development.

14. Discuss in detail on user interface design.

15. Explain the following :

    (a) Black box testing

    (b) Software reviews.

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DE–6418 **304**

DISTANCE EDUCATION

M.C.A./M.C.A (Lateral) DEGREE EXAMINATION, MAY 2013.

Third Semester

VISUAL PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are menus? How to create them?

2. Explain the file system controls with examples.

3. Discuss the various methods and events for the Data control.

4. Write short notes on VBx controls.

5. What is document view architecture?

6. Explain the MFC interface elements.

7. Explain dynamic linking libraries.

8. Write short notes on ODBC.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the function of dialog box with example.

10. Discuss the various methods and events for the data control.

11. Explain how errors are detected and debugged in VB.

12. Explain about mouse event and dialog boxes in VC++ with examples.

13. Explain SDI and MDI.

14. Discuss about object linking and Embedding.

15. Define Active X data object. What are the advantages of ADO?

DE–6419 **401**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Journalise the following transactions of M/S Harish Nandha & Co.

1 March 2012	Purchased goods from Bala Rs. 2,00,000
4th March 2012	Sold goods to Arun Rs. 40,000 and allowed a discount at 5%
8th March 2012	Paid cash to Bala Rs. 1,00,000 and he allowed a discount of Rs. 500
10th March 2012	Paid advertising expenditure of Rs. 20,000

2. Explain the items of sources and application of funds.

3. What do you understand by ratio analysis? Explain its significance and utility.

4. Prepare a cost sheet from the following :

	Rs.
Direct labour	15,000
Direct material	45,000
Direct expenses	10,000

	Rs.
Supervisor salary (factory)	20,000
Supervisor salary (office)	10,000
Power (60% factory)	40,000
Salesmen commission	2,000
Advertisement	18,000
Sales	1,75,000

5. Write a detailed note on break even analysis.

6. Discuss the objectives and limitations of budgetary control.

7. A project cost Rs. 5,00,000 and yields annually a profit of Rs. 80,000 after depreciation at 12% per annum, but before tax of 50%. Calculate payback period.

8. Estimate the working capital requirements from the given information.

	Rs.
Annual sales	7,80,000
Percentage of net profit to cost	25%
Credit allowed to debtors	8 weeks
Credit allowed to creditors	4 weeks
Stock	6 weeks

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Prepare trading and P and L a/c from the given information of M/S Jothi trading Corporation for the year ended 31 March 2012.

	Rs.		Rs.
Opening stock	26,240	Discount received	120
Sales	89,470	Coal and water	720

	Rs.		Rs.
Purchases	33,700	Salaries	4,670
Carriage inwards	4,370	Bank charges	140
Wages	21,470	Rent	600
Depreciation :		Bad debts	680
Building	625	Interest on drawings	300
Plant	1,427	Closing stock	29,390
Furniture	125	Bank charges	140
Interest on capital	4,500	Trade charges	600
		Fire insurance	365

10. Determine the value of

    (a) Current assets

    (b) Current liabilities

    (c) Liquid assets

    (d) Stock from the given information.

    Current ratio 2.5

    Liquid ratio 1.5

    Working capital Rs. 90,000.

11. Prepare a cash flow statement from the given information of Saravanan Ltd.

Liabilities	2010 Rs.	2011 Rs.	Assets	2010 Rs.	2011 Rs.
Share capital	19,000	18,000	Land and building	6,200	5,000
P and L appropriation A/c	3,300	2,900	Patent rights	900	800
Trade creditors and B/P	7,800	6,400	Trade debtors	19,000	15,500
			Cash	4,000	6,000
	30,100	27,300		30,100	27,300

12. From the given information calculate

    (a) Material cost variance

    (b) Material price variance

    (c) Material usage variance of product X.

    The product requires 20 kg of material at Rs. 4 per kg. The actual consumption of material for the manufacturing of product came to 24 kgs of material at Rs. 4.50 per kg.

13. Prepare a flexible budget from the given information and determine the overhead rates at 70% and 90% plant capacity.

	@ 80% capacity (Rs.)
Variable overheads :	
Indirect labour	1,20,000
Stores including spares	40,000
Semi variable overheads :	
Power (30% fixed)	2,00,000
Repair and maintenance (40% variable)	20,000
Fixed overheads :	
Depreciation	1,10,000
Insurance	30,000
Salaries	1,00,000
	6,20,000
Estimated direct labour hours	1,24,000 hrs

14. Explain the objectives of financial management.

15. Discuss the factors that determine the working capital requirements.

DE–6420 **402**

DISTANCE EDUCATION

M.C.A/M.C.A (Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is CGI? Explain.

2. What is meant by event driven programming?

3. Explain the purpose of HTML.

4. What is a Plug-in? Explain.

5. What is the use of <Frame>?

6. Explain the 'if' statement in Java script.

7. Explain the use of PERL.

8. Explain the term : Web graphics.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the features of Java Language.

10. Discuss the control structures of VB Script.

11. Explain any ten tags in HTML.

12. What is an active X Control? Explain its creation and usage.

13. Discuss the various graphics functions. How these functions are used to create animation. Explain.

14. Discuss the components of a distributed system.

15. Explain :

    (a) Client Server Model

    (b) TCP/IP.

DE–6421 **403**

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is Encapsulation? Explain its use with examples.

2. Explain the various kinds of relationships in objects.

3. Explain measuring quality of an Abstraction.

4. What are the features of UML?

5. Explain the purpose and analysis of design.

6. Differentiate between static and dynamic semantics.

7. Explain the advanced concepts in Interaction diagram.

8. Explain state diagrams with examples.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the evolution of object model.

10. Explain any five kinds of relationships among classes.

11. Explain identification of key abstractions and mechanisms.

12. Describe the overview of the view Process.

13. Explain class categories related to class diagrams with examples.

14. Explain the advanced concepts in class diagrams.

15. Explain the features in UML programming.

DE–6422 **404**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the following: compilers, interpreters, assemblers, cross compilers.

2. Explain the method of constructing an NFA from a regular expression. Give an example.

3. With an example, explain the method of constructing operator precedence parsing table for a grammar.

4. Discuss using ambiguous grammars.

5. Explain constructing LALR sets of items.

6. Explain three-address codes. Give examples.

7. Explain symtactic-phase errors.

8. What is peephole optimization? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Construct a minimum state DFA to accept all strings represented by the regular expression ((a+b) (a+b))*.

10. Explain the following:

    (a) Structure of a compiler

    (b) Compiler-writing tools.

11. Construct predictive parsing table for the following grammar:

    $E \rightarrow E+T/T, T \rightarrow T*F/F,\ F \rightarrow (E)/id.$

    Also explain the method of parsing a given sentence using the table.

12. Explain how syntax-directed translation scheme is used for generating intermediate code. Give examples.

13. Explain the data structures used for implementing symbol tables.

14. Discuss loop optimization.

15. Explain the following:

    (a) Problem in code generation

    (b) Simple code generator.

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DE–6423 **501**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the concepts of Network operating system in detail.
2. Discuss the communication protocols for RPC.
3. Explain the different types of addressing used in a client server model.
4. Explain how logical clocks are implemented.
5. Explain about distributed deadlock detection.
6. Explain the implementation of distributed file system structure.
7. Explain Ethernet and WiFi.
8. Explain how the thread packages are implemented.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss about design issues distributed file system.
10. Discuss about advantages of distributed system.

11. Explain about the implementation of atomic transaction.

12. What are the algorithms available to control concurrency? Explain.

13. Explain client server communication.

14. Explain Cryptographic algorithm in detail.

15. Explain multicast communication.

DE–6424 **502**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Write a note on cascading style sheets.

2. What is SAX? When to use it?

3. What do you understand by Java Bean? What are its advantages?

4. Write a note on Http Servlet Request Object.

5. What is struts? Why are the used? What are the three components of struts?

6. Explain the structure of a JSP Page.

7. What do you mean by scripting elements in JSP? List out its types.

8. Narrate on the steps involved in creating a JDBC program.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Write an essay on steps involved in reading XML data through DOM.

10. Give an account on different types of EJB.

11. Explain the life cycle of a Servlet.

12. How cookies are used in session tracking mechanism?

13. Give a detailed note on the implicit JSP objects.

14. Explain the steps to install Tomcat Server and its testing environment.

15. With a suitable example, explain how to access a database from JSP Page.

DE–6425 **503**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

.NET FRAMEWORKS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare and contrast .NET with other programming languages.

2. Define interoperability. How .NET supports interoperability? Explain.

3. Write a note on assembly.

4. Explain any four string handling functions in .NET.

5. Define Thread. Explain the various states of threads.

6. Explain the steps involved in creating Windows forms using VS .NET IDE.

7. Write a note on datasets, schemas and XML.

8. How do you use web services for remoting over the internet? Explain.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain the architecture of .NET Framework.

10. What happens when you compile and run a program in .NET?

11. Write a note on System .Object class in .NET.

12. Write a note on :

    (a) The .NET base class library.

    (b) Managed code and its benefits.

    (c) Intermediate language and its features.

13. Define ADO .NET. List the characteristics of ADO .NET.

14. What do you mean by exposing web services and consuming web services? Explain.

15. Write a program in any .NET language for scheduling threads.

———————

DE–6426 **504**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a short note on Data mining versus query tools.
2. Explain briefly about OLAP tools.
3. Explain about load manager in process architecture.
4. Describe the process of designing the starflake schema.
5. Discuss briefly about Network hardware.
6. Explain on the performance impact of security.
7. Write a short note on tuning queries.
8. Explain briefly about testing the operational environment.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write short notes on the following :
   (a) Information and production factor
   (b) Data mining in marketing.

10. Explain about the preliminary analysis of the data set using traditional query tools.

11. Explain in detail about data warehouse schemas.

12. Describe the steps and guidelines effective in determining facts from dimensions.

13. Explain in detail about hardware partitioning.

14. Describe the server hardware with its architecture options.

15. Explain the process of estimating the load in capacity planning.

DE–6427 **601**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. What is meant by routing?

2. Write short notes on SDMA.

3. Write notes on Roaming.

4. What is meant by Hand over? Explain with examples.

5. Explain about encapsulation.

6. Explain the various features of mobile IP.

7. What is selective retransmission? Explain briefly.

8. Define wireless mark-up language (WML).

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain the various services of UMTS in detail.

10. Write notes on FDMA and TDMA.

11. Describe in detail about location management.

12. Write notes on :
    (a) Reverse Tunneling
    (b) Adhoc Networks.

13. Explain in detail about snooping TCP.

14. Write notes on session protocol and transaction protocol.

15. Explain the overview of broadcast systems.

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DE–6428 **602**

DISTANCE EDUCATION

M.C.A./M.C.A (Lateral) DEGREE EXAMINATION, MAY 2013.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. List the CORBA services and state the difference between CORBA and RMI.

2. Discuss on the common Architecture of an e-commerce website.

3. Discuss on creating a CSS and also state the prons and cons of using CSS.

4. Discuss on the Data management services.

5. Discuss in brief on the security issues of Beans.

6. Write a program to generate the login control using ASP.NET

7. Create a UDDI business registry and perform search from the registry.

8. Discuss on the validation controls of ASP.NET

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Discuss on the MVC Architecture in detail.

10. Applying RMI, perform shopping applications through cart.

11. Discuss on Session Beans in detail.

12. Discuss in detail the structure of the object adapter.

13. Write a program to create a student information system project to view the records using Grid view and detail view using ASP.NET.

14. Design a web page involving multi view or wizard control.

15. Discuss on the security management of ASP.NET.

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DE–6429 **603**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss in detail on game playing.
2. Give the semantic net representation.
3. Discuss on depth first and breadth first search with examples.
4. Explain chaining with example.
5. Discuss on any one application of Graphs.
6. Discuss on the utility theory.
7. How are various types of knowledge imparted in learning? Discuss with examples.
8. Discuss on the future of AI.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail on the constraint satisfaction problem.
10. Discuss on any two problem solving technique applying it to game playing.

11. Discuss in detail the first order logic and first order inference.

12. Discuss in detail Bayes rule with application.

13. Explain decision networks in detail.

14. Explain reinforced learning in detail.

15. Discuss in detail Formal grammar with examples and application.

DE–6430 **604**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Mention the usage of Free Open Source Software.

2. How Ethernet can be used for Internet connection?

3. Discuss the OS booting up process.

4. How to set up proxy services?

5. List the need for net filter.

6. How to construct make files?

7. Discuss the usage of GNU debugging tools.

8. Explain the concept of GtK + Programming.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the process of installing a GNU/Linux distribution.

10. Mention the basics of TCP/IT networking.

11. How to set up email servers?

12. Briefly discuss on :

    (a) ccp

    (b) gcc

    (c) g++.

13. How to use gdb to debug programs?

14. Describe the architecture for X-windows server.

15. Give the importance of Python programming.

DE–6647 **106**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

Break up of marks
Record Note Book : 10
Table creation : 15
Query design : 30
Entering/debugging : 30
Result : 15

1. (a) A private company provides salary to its employees in the following :

DA – 43% of bpay

HRA – 7% of bpay

PF – 12% of bpay

LIC – subscribed by an amount

Create a master table of 10 records

Master (emp_name, emp no, designation, bpay, DA, HRA)

Transaction (emp-no, PF, LIC)

Write a SQL procedure to prepare a payable with all details including grosspay and netpay.

(b) Create the following table and insert atleast 10 records.

Marks (st_number, subject, mark, result)

Write a PL/SQL procedure to display the records in the table by subject wise.

2. (a) Create the following tables and insert atleast 10 records.

Book (book-id, title, author-id, publisher-id, category)

Author (author-id, name)

Publisher-id (publisher-id, name, year)

Write SQL queries to do the following:

(i) display the books in the category 'computer'

(ii) count the number of books on each category

(iii) list the books published after 2010

(iv) list the books names by the same author.

(b) Consider the following table :

Transaction (item-no, tr-date, tr-type, quantity) if tr-type is 'S' for sales and 'P' for purchase

Create a procedure to receive an item number and date and list all the transaction connected with these two.

---------- Cut here ----------

3. (a) Consider the following tables and write queries candidate (cand-code, age, party-name, votes polled)

constituency (con-code, total voters)

Winner (con-code, cand-code)

(i) List all the candidates with their votes in a particular constituency.

(ii) List the winners in all the constituencies.

(iii) List the winner of a particular party.

(b) Consider the following table and write queries marks (student-no, mark, result)

(i) Write procedure to update result as 'pass' if mark>50 other wise 'FAIL'

(ii) List the students who failed

(iii) List the student no who got first mark

(iv) List the student who got more than 80.

4. (a) Create the following table and insert atleast 10 records by using NULL for net-reading and bill-amount.

EB (cust_id, cust_name, pre_reading, cur_reading, net_reading, bill_amount)

Calculate the net-reading and bill-amount and up date in the table.

net-reading	amount per unit
1–200	Rs. 2.00
200–400	Rs. 3.00
400 & above	Rs. 5.50

(b) Write a SQL procedure prepare pay slip for an employee of an organisation.

(Assume your own data and format)

---------- Cut here ----------

5. (a) Create the following table for 10 students and insert only stud-id, name, mark 1, mark 2 and mark 3 student (stud_id, name, mark 1, mark 2, mark 3, total, average, result).

Write SQL procedure to calculate total average and result of each student and update the same. Set the result as "PASS" 16 average is > 50 otherwise "FAIL".

(b) Create the table and insert atleast 10 records.

Employee (emp_code, emp-name, dept, bp)

Prepare a salary bill of the all employees with BP, DA, HRA, PF and NET PAY.

DA = 50% of BP

HRA = 10% of BP

PF = 12% of BP

NETPAY = BP + DA + HRA – PF.

DE–6648 **205**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Record : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiner has to choose ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ problem to draw a ball and make the ball movement from left to right and vice-versa.

   (b) Write a C++ program to draw a straight line using simple DDA algorithm.

---------- Cut here ----------

2. (a) Develop a C++ program to draw an ellipse.

   (b) Develop a C++ program to draw a circle. Divide into four quadrants and apply different colors to each quadrant.

3. (a) Write a C++ program to implement the polygon clipping algorithm with graphics functions.

   (b) Develop a C++ program to draw a ball and perform animation on it.

---------- Cut here ----------

4. (a) Write a C++ program to draw a car with rectangle, circles etc. Also make the car to move from one direction to other.

   (b) Develop a C++ program to translate the triangle to the right of its original position by 5 units.

---------- Cut here ----------

5. (a) Write a C++ program to draw a straight line using simple DDA algorithm.

   (b) Develop a C++ program to draw non-linear objects arcs and curves.

---------- Cut here ----------

6. (a) Write a C++ program to manipulate the graphical object using SHADING technique.

   (b) Write a C++ program to rotate a two dimensional triangle by 45° in clockwise and anti-clockwise direction.

---------- Cut here ----------

7. (a) Write a C++ program to draw a straight line using Bresenham's algorithm.

   (b) Develop a C++ program to draw a square and draw a circle within the square having diameter as the side's value.

8. (a) Write a C++ program to draw a triangle with three sides of equal and translate it to the right by 10 units from the origin.

(b) Write a C++ program to draw a ball and make the ball to move in sine curve form.

---------- Cut here ----------

9. (a) Develop a C++ program to draw a circle and three concentric circles.

(b) Write a C++ program to draw a straight line using DDA technique.

---------- Cut here ----------

10. (a) Develop a C++ program to draw a cube with different colors on each face.

(b) Write a C++ program to draw non-linear objects curves and arcs.

DE–6649 **206**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

UNIX AND SHELL PROGRAMMING – LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Each candidate should select ONE question using lot system.

Each subdivision carries 50 marks.

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

1. (a) Write a shell program to perform the following with respect to files.

   (i) Counts the number of words is a file.

   (ii) Name of a person is present in two files.

   (b) Write a shell program to find how many days remaining in a current year.

2. (a) Write a menu driven shell program for the following :

   (i) To check if a file is writeable.

   (ii) Exit from Unix.

   (b) Write a shell program to accept a number from 0 to 9 and display the same number in words.

---------- Cut here ----------

3. (a) Write a shell program to perform the following :

   (i) Enter any ten numbers in a file.

   (ii) Sort the contents in Ascending order.

   (b) Write a shell program to accept a string and check whether it is a palindrome or not.

---------- Cut here ----------

4. (a) Write a shell program to display all the names of the files starting with the letters the current directory.

   (b) Write a shell program to generate a Fibonacci series.

---------- Cut here ----------

5. (a) Write a shell program for the following :

   (i) List the files in date order.

   (ii) Sort the files in descending order.

   (b) Write a shell program to check whether given number is even or odd using positional parameters.

6. (a) Write a shell program to arrange the names in alphabetical order with swapping done in a separate function.

   (b) Write a shell program to reverse a string.

---------- Cut here ----------

7. (a) Write a shell program to check if a number is positive, negative or zero.

   (b) Write a shell program to find the following objects :

      (i) Null string

      (ii) Not a null string.

DE–6650 **305**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Third Semester

NETWORK LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiners should select and give ONE question to each candidate by lot system.

Break-up of marks :

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a Java program to implement of socket programming using TCP.

   (b) Write a Java program to read file from the remote system.

---------- Cut here ----------

2. (a) Write a Java program to implement of ECHO server and client using TCP.

   (b) Write a Java program to send message from one system to another system.

3. (a) Write a Java program to implement of remote command execution using TCP.

   (b) Write a Java program to add two numbers using RMI client, server technique.

---------- Cut here ----------

4. (a) Write a Java program to implement of socket programming using UDP.

   (b) Write a Java program, find factorial of given number using RMI client, server technique.

---------- Cut here ----------

5. (a) Write a Java program to implement of PING server using RAW socket.

   (b) Write a Java program to setup time server and time client.

---------- Cut here ----------

6. (a) Write a Java program to implement of Remote Method Invocation (RMI).

   (b) Write a Java program to setup a chat server and chat client.

---------- Cut here ----------

7. (a) Write a Java program to implement Date and Time Display from local Host to server using TCP.

   (b) Write a Java program to ENCRYPT given word.

8. (a) Write a Java program to create sockets for sending and receiving Data.

(b) Write a Java program to display the user and their IP addresses logged in the network.

DE–6651 **306**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Third Semester

VISUAL PROGRAMMING – LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Students has to answer ONE question by lot system.

1. (a) Develop an Applications to create menu control Arrays at run time.

   (b) Write a visual C++ Program to find the power of a number.

---------- Cut here ----------

2. (a) Develop an application to illustrate the drag and drop operations.

   (b) Write a visual C++ Application program using MFC that display current mouse co-ordinates in status bar.

3. (a) Develop an application for a calculator performing addition, multiplication, division, subtraction and percentage operations using command buttons.

(b) Write a visual C++ application program using MFC that creates a new font style.

---------- Cut here ----------

4. (a) Using a FlexGridControl write a program that calculate the additions, subtraction, multiplication and Division of numbers ranging from 1 to 12.

(b) Create a windows using C++ sources files.

---------- Cut here ----------

5. (a) Write a VB program using Active X control to create a text box that accepts only numeric value. It should include back color, fore color and text.

(b) Writ a visual C++ program to calculate a factorial of a given n number.

DE–6652 **405**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

INTERNET PROGRAMMING LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Students has to answer ONE question by Lot system.

Each subdivision carries 50 marks.

1. (a) Write a java applet program to display a filled red color circle within an ellipse.

   (b) Write a VB script to check the given user name and pass word only three attempt.

---------- Cut here ----------

2. (a) Write Java Script to create a color palet and display "JAVA SCRIPT" text in the color choosen from the color palat.

   (b) Write a Java applet program to display the menu file includes new, open save and exit.

---------- Cut here ----------

3. (a) Develop a single page advertisement for super market shop.

   (b) Write a VB script to create a calender for given month and year.

4. (a) Develop a photo gallery web page with thumb images. When click the thumb image show a original size photo in new page.

   (b) Write a Java program using JDBC to display your subject paper title for a given subject code.

---------- Cut here ----------

5. (a) Create a web page to display the course details of your department.

   (b) Write a VB script to do the following :

      (i) Display the current time and date.

      (ii) Find the age for a given date of birth.

      (iii) Find the difference between the two dates.

DE–6653 **406**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S)(Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

COMPILER DESIGN – LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break up of marks :

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging/Execution	: 35
Result	: 10
Total	: 100

1. Write a program to convert an infix expression into postfix notation and three address code. Check your program with the following expression

$$a + b * (c - d)/(e + f).$$

---------- Cut here ----------

2. Write a program to construct canonical collection of $LR(0)$ items for the following grammar

$$E \to E + T / T,\ T \to T * F / F,\ F \to (E)/id.$$

3. Write a program to construct predictive parsing table for the following grammar

$$E \to E + T / T,\ T \to T * F / F,\ F \to (E) / id.$$

---------- Cut here ----------

4. Write a program to draw transition diagram for DFAs to accept strings represented by the following regular expressions

(a) $((a+c)(b+a)(a+b))^*$

(b) $(0+1)*011$.

---------- Cut here ----------

5. Write a program to construct $LR(0)$ parsing table for the following grammar

$$S \to CC,\ C \to Cc / d.$$

---------- Cut here ----------

6. (a) Write a program to find keywords in a source program.

(b) Write a program to convert infix expression into postfix notation.

---------- Cut here ----------

7. (a) Write a program to construct leftmost and rightmost derivations for the following sentences from the given grammar

(i) $id * (id + id)$

(ii) $(id + id) * (id + id)$

The grammar is

$E \to E + T / T,\ T \to T * F / F,\ F \to (E) / id$.

(b) Write a program to remove comment lines from a source program.

8. (a) Write a program to identify constants to a source program.

(b) Write a program to find LEADING and TRAILING sets for the following grammar

$$S \rightarrow AbB,\ A \rightarrow aBb,\ B \rightarrow a / Bba\,.$$

DE–6654 **505**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

.NET LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Students has to answer One question by lot system.

Each subdivision carries 50 marks.

1. (a) Create a simple application that retrieves information from student table and display it in a simple label control when the uses select the student name using a drop-down list box.

   (b) Create a simple application that retrieves information from student table and display the information from each field in a separate text boxes when the uses select the student name using a drop-down list box.

———————————— cut here ————————————

2. (a) Create a simple application that update the information in the text boxes to the corresponding fields in student table when the user clicks the update button.

   (b) Create an application that retrives and display the information of bank customer from multiple tables. When the user select the customer's account number from the drop-down list box.

3. (a) Develop an application that displays the information of all students in grid view control excluding the address field with selection, edition and sorting option.

   (b) Develop a simple file browser program that allows the uses to see information about any file in any directory in the current drive.

———————————————— cut here ————————————————

4. (a) Develop a simple e-mail application in ASP.NET.

   (b) Create an advertisement link to another page using AdRotator.

———————————————— cut here ————————————————

5. (a) Write a web application to develop user registration form (use validation control).

   (b) Develop a splash screen which will show a message to passenger regarding the arrival and departure of trains for every 2 minutes.

———————————————— cut here ————————————————

6. (a) Develop a web service than has a ASP.NET client.

   (b) Develop a web service to fetch data from a table and send it across to the client.

———————————————— cut here ————————————————

7. (a) Create a web page to add, edit, delete and view the information of students.

   (b) Develop an application to generate a crystal report from an existing student database.

DE–6655 **506**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Fifth Semester

WEB TECHNOLOGY – LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system

Break up of marks

Record Note Book : 5

Algorithm/flowchart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Create a HTML frames and split the frame Horizontal.

   (b) Write an ASP program to prepare students mark list.

---------- Cut here ----------

2. (a) Write a servelet program to display "Good morning".

   (b) Create a image animation using Java script.

3. (a) Create a web page in the format of front page of arts college information using text links. Align the text with colors.

   (b) Write an XML document to display your's name, Date of Birth, Address and phone number. Write an XSL style sheet and attach to the XML document. Validate the document using DTD.

---------- Cut here ----------

4. (a) Write a program to display day, month and year using serverlets.

   (b) Write an JSP program to prepare telephone bill using Java script.

---------- Cut here ----------

5. (a) Write JSP program to display worker Id, worker name, worker designation and salary using JDBC.

   (b) Write a servelet program to display your college name.

---------- Cut here ----------

6. (a) Write an XML document to display your favorite Indian leaders. Write an XSL style sheet and attach that to the XML document validate the document using XSD.

   (b) Write a servelet program to create a chatting application.

---------- Cut here ----------

7. (a) Create a HTML frame using various controls.

   (b) Write a CSS code which can display the different types of cursor.

8. (a) Write a servelet program to display your district name.

(b) Write a Java script code to flip between two different fonts using on mouse over and on mouse out event.

DE–6656 **605**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the LOT system and give it to the student.

1. (a) Write a server side PHP program that displays marks, total, grade of a student in tabular format by accepting user input for name, number and marks from a HTML form. (45)

   (b) Write a PHP program to upload a file to the server. (45)

   (c) Record. (10)

---------- Cut here ----------

2. (a) Write a PHP program that adds products that are selected from a web page to a shopping cart. (45)

   (b) Write a PHP program to create a director and to read contents from the directory. (45)

   (c) Record. (10)

3. (a) Write a PHP program to access the data stored in a my sql table. (45)

   (b) Write a shell program to find the details of an user session. (45)

   (c) Record. (10)

---------- Cut here ----------

4. (a) Write a PHP program interface to create a database and to insert a table into it. (45)

   (b) Write a shell program to change the extension of a given file. (45)

   (c) Record. (10)

---------- Cut here ----------

5. (a) Write a PHP program using classes to create a table. (45)

   (b) Create a my sql table and execute queries to read, add, remove and modify a record from that table. (45)

   (c) Record. (10)

DE–6657 **606**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Sixth Semester

MULTIMEDIA TOOLS — LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

Break-up of marks :
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Implement a program in Flash for bouncing a ball.

   (b) Animate your name with different colours. Make the characters to Blink.

---------- Cut here ----------

2. (a) Develop an animation to lighten the candle and show its shadow effect.

   (b) Animate your own text to create the Horizontal flipping effect.

3. (a) Create any two different shapes animate the shapes slowly appear and disappear one after another.

   (b) Implement a program in Maya to create class room using Rendering.

---------- Cut here ----------

4. (a) Implement a program in Flash to develop Masking Animation.

   (b) Develop a Home Page using Dream viewer for the Mobile Showroom.

---------- Cut here ----------

5. (a) Develop an animation in Flash to glow the bulb in different colours.

   (b) Create a Cartoon with Laughing Face.

---------- Cut here ----------

6. (a) Develop an animation to show a growth tree.

   (b) Implement a program to create a Maze using Maya.

---------- Cut here ----------

7. (a) Implement an animation to fly a bird in the Forest.

   (b) Animate your own text to create the Rotating effect in the clock wise direction and print in Reverse format.

---------- Cut here ----------

8. (a) Implement a program for animating flight between two mountains using Flash.

   (b) Develop a web page using Dream viewer for your favourite Engineering College.